

HCSM-21

The Time Theorem

Complex Time from the Five Layer Loop, the Topological Derivation of the Floor Function, the Processor as Projection onto Nonzero Real Time, and the Landauer Selection of T_{DME}

Hodge Complex Standard Model

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Abstract

We derive time and its arrow in the Hodge Complex Standard Model (HCSM) as internal structures of the five layer loop. The loop operator

$$M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}$$

acts diagonally on the 14-mode multiplet with polar form $M_{\text{loop}}(m) = r_m e^{i\theta_m}$. The complex time of mode m is

$$\tau_m = \theta_m + i \ln r_m = \omega_m T_{\text{DME}} - i(-\ln r_m),$$

where ω_m is the winding-generator frequency, $T_{\text{DME}} = 1/(2\gamma)$ is the DME relaxation time, and r_m is the per-pass amplitude. The real part θ_m is the clock; the imaginary part $-\ln r_m > 0$ is the arrow increment. The processor T is exactly the projection onto modes with $\text{Re}(\tau) > 0$.

Four structural bridges are established. (i) The floor function in the Core perturbation is the winding number of the DME-Core interface phase over one loop pass; it is not an ansatz. (ii) Dimensional homogeneity of τ_m is proved. (iii) The exact ratio $|M_A|/|M_B| = 3 = d - 1$ is a consequence of the closed-form T_{DME} from HCSM-54. (iv) The loop-pass duration selection is closed: T_{DME} is the unique dissipative scale in HCSM, and the processor T at the deterministic Landauer bound requires dissipation. The coherent T_{beat} is rejected because it carries no entropy transfer.

Derivation status summary. *Derived:* loop operator diagonalization; complex time closed form; topological floor derivation; dimensional homogeneity; processor identity; arrow of time; exact ratio $|M_A|/|M_B| = 3$; **loop-pass duration** $T_{\text{DME}} = 1/(2\gamma)$. *Disclosed inputs:* stability coefficient γ (HCSM-54); EFT parameters w and $\zeta H/\rho$ (HCSM-56). *Open:* identification of continuous time $t = kT_{\text{DME}}$ from a deeper principle; physical conversion of T_{DME} to seconds.

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1 Introduction

1.1 The time problem

The Standard Model has no fundamental arrow of time. HCSM supplies a closed five-layer loop with preferred direction and a dissipative operator. Time is a readout of the loop, not an input.

Two aspects emerge:

- (1) Clock time — the phase accumulated by M_{loop} per pass.
- (2) Arrow time — the directed decay of M_{loop}^k because $|M_j| < 1$.

1.2 Summary of results

- (1) (**Theorem 8.2**). M_{loop} is diagonal with $\tau_m = \omega_m T_{\text{DME}} + i \ln r_m$; closed forms $\tau_A = 2.0121430664919213 - 0.5521962501528914i$, $\tau_B = 6.0364291994757650 - 1.6508085388210012i$, $\tau_C = 0$.
- (2) (**Theorem 4.3**). Floor function = winding number of DME-Core interface.
- (3) (**Proposition 8.3**). Dimensional homogeneity of τ_m .
- (4) (**Theorem 9.1**). $T = \Pi_{\text{Re}(\tau) > 0}$.
- (5) (**Theorem 11.1**). Arrow of time.
- (6) (**Theorem 12.1**). Loop-pass duration selection: $T_{\text{DME}} = 1/(2\gamma)$.

1.3 Organization

Sections 2–7 develop the five layers. Section 8 defines complex time. Section 9 proves the processor identity. Sections 10–12 close the ratio, arrow, and Landauer selection. Section 13 states the Time Theorem.

2 Setup: the five layer loop

2.1 State spaces

Layer	State space	Operator
1: FDLM/WIN	$\mathbb{C}^{14}$ mode vectors	Hermitian (unitary)
2: DME	$\epsilon \in \mathbb{R}$	scalar constraint
3: Core	$\epsilon(\lambda)$	gradient flow
4: EFT	field config	damped wave equation
5: Processor T	$\mathbb{C}^{14}$ projected	orthogonal projection

2.2 The loop operator

$$M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}, \quad M_{\text{loop}}(m) = r_m e^{i\theta_m}. \quad (1)$$

The five layers are algebraically composed, not temporally sequential. Time is read from the eigenvalues of the composition.

3 Level 1 — FDLM/WIN: winding phase

$H_{\text{wind}} = i \sin((\pi/8)M)$ with $\text{spec}(H_{\text{wind}}) = \{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}$. The beat period is $T_{\text{beat}} = 2\pi/(2\sqrt{2}) = \pi/\sqrt{2}$.

The 14 modes partition into:

- Class A: $\omega_A = 2\sqrt{2}$, 3 modes.
- Class B: $\omega_B = 6\sqrt{2}$, 7 modes.
- Kernel: $\omega = 0$, 4 modes.

The ratio $\omega_B/\omega_A = 3$ is exact.

4 Level 2 — DME: time quantum and topological floor

4.1 Time quantum

$\gamma = c_2/I(2) = 0.70283946799\dots$ (HCSM-54); $T_{\text{DME}} = 1/(2\gamma) = 0.7114000035\dots$

4.2 Phase handoff

$\Phi_{1 \rightarrow 2}(m) = \omega_m T_{\text{DME}}$. For class A: $\Phi_A = 2 + \delta_{\text{edge}}$; for class B: $\Phi_B = 6 + 3\delta_{\text{edge}}$, $\delta_{\text{edge}} = \sqrt{2}/\gamma - 2 = 0.0121430664919\dots$

4.3 Topological floor derivation

Definition 4.1 (DME-Core interface). $\gamma_m : S^1 \rightarrow U(1)$, $\gamma_m(s) = e^{2\pi i \omega_m T_{\text{DME}} s}$, $s \in [0, 1)$.

Lemma 4.2 (Lift and winding). *The lift $\widetilde{\gamma}_m(s) = \omega_m T_{\text{DME}} s$ crosses integer values $\lfloor \omega_m T_{\text{DME}} \rfloor$ times per pass.*

Theorem 4.3 (Core perturbation quantization). *The unique minimizer of the DME-Core functional $J[\epsilon_0]$ on $\delta_{\text{edge}} \mathbb{Z}_{>0}$ is $\epsilon_0(m) = n_m \delta_{\text{edge}}$, $n_m = \lfloor \omega_m T_{\text{DME}} \rfloor$.*

The floor function is not an ansatz — it is the winding number of the DME-Core phase over one pass.

5 Level 3 — Core: the arrow of time

$d\epsilon/d\lambda = -2\gamma\epsilon$; Lyapunov $\mathcal{L}[\epsilon] = \gamma\epsilon^2$, $d\mathcal{L}/d\lambda = -4\gamma^2\epsilon^2 < 0$. Strictly monotone. Master conditions M1 (monotonicity) and M2 (irreversibility) satisfied.

6 Level 4 — EFT: damping

$\phi'' + \Gamma_{\text{eff}}\phi' + (k^2/a^2 + m_{\text{eff}}^2)\phi = 0$. From HCSM-56: $w = -1 - (1 - \gamma)^2/(2\pi)$, $\zeta H/\rho = (1 - \gamma)^2/(6\pi)$. Decay factor per beat: $D_{\text{factor}} = \exp(-(\zeta H/\rho)T_{\text{beat}}) = 0.994810128\dots$

7 Level 5 — Processor T

$T = P_{\text{mesh}} + P_{\tau}$; $T^2 = T$; $T^\dagger = T$; $\text{rank}(T) = 10$; kernel $\{\text{dark}, W, Z, H\}$.

8 The complex-time operator

Definition 8.1 (Complex time). $\tau_m = \theta_m + i \ln r_m = \omega_m T_{DME} - i(-\ln r_m)$.

Theorem 8.2 (Closed form).

$$\begin{aligned}\tau_A &= 2.0121430664919213 - 0.5521962501528914i, \\ \tau_B &= 6.0364291994757650 - 1.6508085388210012i, \\ \tau_C &= 0.\end{aligned}$$

8.1 Loop-decay decomposition

$$r_m = [|w + 1|/\epsilon_0(m)]_{\text{Core}} \cdot [D_{\text{factor}}]_{\text{EFT}}.$$

Proposition 8.3 (Dimensional homogeneity). *Both θ_m and $-\ln r_m$ are per-pass products of a rate with T_{DME} , and both scale linearly with the Core flow parameter λ .*

Structural identities: $\theta_B/\theta_A = 3$ (exact); $\ln(r_A/r_B) = \ln 3$ (exact).

9 The processor as projection onto nonzero real time

Theorem 9.1 (Processor identity). $T = \Pi_{\text{Re}(\tau) > 0}$. *Modes in $\text{im}(T)$ have $\omega_m \in \{2\sqrt{2}, 6\sqrt{2}\}$, hence $\text{Re}(\tau_m) > 0$; modes in $\ker(T)$ have $\omega_m = 0$, hence $\tau_m = 0$.*

10 The exact ratio $|M_A|/|M_B| = 3$

Theorem 10.1 (Exact ratio). $|M_A|/|M_B| = r_A/r_B = 3 = d - 1$, *independent of $|w + 1|$, δ_{edge} , and D_{factor} .*

11 The arrow of time

Theorem 11.1 (Arrow). *The arrow is the direction of increasing k , equivalently decreasing $\mathcal{L}[\epsilon]$, equivalently increasing $-\ln r_m$.*

12 The Landauer selection of T_{DME}

Prior versions of this paper left the loop-pass duration selection open: both $T_{DME} = 1/(2\gamma)$ (DME relaxation) and $T_{\text{beat}} = \pi/\sqrt{2}$ (Level-1 internal period) are dimensionless time-like quantities. This section closes that item.

Theorem 12.1 (Loop-pass duration). *The loop-pass duration is*

$$T_{\text{pass}} = T_{DME} = \frac{1}{2\gamma} = 0.711400003516965862\dots \quad (2)$$

Proof. The loop contains the processor $T = I - P_C$ (HCSM-13), a Maxwell demon operating at the deterministic Landauer bound. The processor takes the maximally mixed state on the 56-dim sector ($S_{\text{in}} = \log 56$) to the factorized state on $10 + 4$ modes ($S_{\text{out}} = \log 40$), with entropy reduction $\Delta S = \log(7/5) = 0.336472\dots$ nats.

Landauer's principle requires that this entropy reduction be compensated by heat dissipation into the residual bath. Dissipation requires a dissipative rate. Among the HCSM-native scales:

- (i) $T_{DME} = 1/(2\gamma)$ is derived from the gradient flow $d\epsilon/d\lambda = -2\gamma\epsilon$ (HCSM-54). It is the unique dissipative scale in HCSM.
- (ii) $T_{beat} = \pi/\sqrt{2}$ is a coherent standing-wave period of the winding generator. It carries no entropy transfer and no irreversibility.

Because the loop is a measurement, it is irreversible; because it is irreversible, it must dissipate; because it dissipates, its time unit must be the dissipative time T_{DME} . The coherent T_{beat} is rejected. $\square$ $\square$

Remark 12.2 (Tautology of the ratio). $T_{beat}/T_{DME} = \pi\sqrt{2}\gamma = 3.1226334\dots$ is a tautology given the definitions of both scales. It carries no structural information about the selection; the selection is physical, via the Landauer argument above.

Remark 12.3 (Verification: no other dissipative scale). The HCSM-native scales are $T_{DME} = 1/(2\gamma)$ (dissipative), $T_{beat} = \pi/\sqrt{2}$ (coherent), $T_{lattice} = 1/L$ (combinatorial), $T_{site} = 1/N$ (combinatorial). Only T_{DME} carries entropy transfer. Rank 7 is closed.

13 The HCSM Time Theorem

Theorem 13.1 (HCSM Time Theorem). (1) No external time coordinate is assumed.

(2) $M_{loop}(m) = r_m e^{i\theta_m}$, $\theta_m = \omega_m T_{DME}$, $T_{DME} = 1/(2\gamma)$.

(3) $\tau_m = \omega_m T_{DME} - i(-\ln r_m)$.

(4) Discrete HCSM time is the loop count k ; continuous time is $t = kT_{DME}$.

(5) Arrow of time: increasing $k \Leftrightarrow$ decreasing $\mathcal{L}[\epsilon] \Leftrightarrow$ increasing $-\ln r_m$.

(6) $T = \Pi_{\text{Re}(\tau) > 0}$.

(7) Kernel modes $\tau_m = 0$; frozen under M_{loop} .

(8) $|M_A|/|M_B| = 3 = d - 1$.

(9) Floor function = DME-Core winding number.

(10) $T_{pass} = T_{DME} = 1/(2\gamma)$ (Landauer selection).

14 Derivation status

14.1 Derived

- (1) Loop operator diagonalization and complex-time formula.
- (2) Closed forms of τ_A, τ_B, τ_C .
- (3) Multiplicative loop-decay decomposition.
- (4) Topological derivation of the floor function.
- (5) Dimensional homogeneity of τ_m .

- (6) Processor identity $T = \Pi_{\text{Re}(\tau)>0}$.
- (7) Arrow of time from Core Lyapunov.
- (8) Exact ratio $|M_A|/|M_B| = 3$.
- (9) **Loop-pass duration** $T_{\text{pass}} = T_{\text{DME}}$ (**Landauer selection**).

14.2 Disclosed inputs

$\gamma = c_2/I(2)$ (HCSM-54); EFT parameters $w, \zeta H/\rho$ (HCSM-56).

14.3 Open

- (1) Identification of continuous time $t = kT_{\text{DME}}$ from a deeper principle.
- (2) Physical conversion of T_{DME} to seconds.

15 Falsifiable predictions

- (1) τ_A and τ_B exactly as stated; a 10^{-10} deviation falsifies.
- (2) $T = \Pi_{\text{Re}(\tau)>0}$: a mode with $\text{Re}(\tau) > 0$ outside $\text{im}(T)$ (or vice versa) falsifies.
- (3) Arrow direction: any loop mode with $|M_j| > 1$ falsifies.
- (4) Kernel freezing: any nonzero time evolution of $\{\text{dark}, W, Z, H\}$ falsifies.
- (5) Loop-pass duration: any experimental or structural result requiring the pass duration to be T_{beat} falsifies Theorem 12.1.

16 Conclusion

Time in HCSM is not an external coordinate. It is the complex eigenvalue of the loop operator, $\tau_m = \omega_m T_{\text{DME}} - i(-\ln r_m)$. Its real part is the clock; its imaginary part is the arrow. The processor T is the projection onto nonzero real time.

Four algebraic bridges are now closed. First, the floor function is the winding number of the DME-Core interface. Second, the exact ratio $|M_A|/|M_B| = 3$ matches the generation count $d - 1$. Third, dimensional homogeneity of complex time. Fourth, the loop-pass duration T_{DME} is selected by Landauer irreversibility; Rank 7 of the HCSM blocker list is closed.

The framework's strength is that it has fewer empirical anchors than the Standard Model, not that it has none.

Physics is what the machine cannot cancel. The Hodge complex is the machine.

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A Reproducible verification

The companion script verifies:

1. Loop operator diagonalization.
2. Closed forms of τ_A, τ_B, τ_C .
3. Multiplicative decomposition.
4. Topological floor derivation.
5. Dimensional homogeneity.
6. Processor identity.
7. Arrow of time.
8. Exact ratio = 3.
9. $T_{\text{DME}} = 1/(2\gamma)$ at machine precision.
10. Landauer bound $\Delta S = \log(7/5)$.

B Web Crawler Metadata

B.1 Dublin Core

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DC.creator: Preschutti, Stanley

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B.2 BibTeX

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  title = {{HCSM-21}: The Time Theorem},  
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  abstract = {We derive time and its arrow in HCSM as internal  
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    selected by Landauer irreversibility:  $T_{pass} = T_{DME} = 1/(2 \gamma)$ .  
    Rank 7 closed.}  
}
```